CRYPTOGRAPHY NETWORK
SECURITY

ASSIGNMENT - II

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SET - II

1) Blowfish - S-box 1 in the F Function: 98/100

Given input

01101001011010010101001 = decimal 105

* There is a small error in the question Blowfish's function normally takes a 32-bit input. divided into four 8-bits parts

$$x = a \,||\, b \,||\, c \,||\, d$$

for the given repeated 8-bit value 01101001 (0x69) we can treat it as

* a = 01101001 = 105
* b = 01101001 = 105
* c = 01101001 = 105
* d = 01101001 = 105

* So, S box 1 is accessed at entry 105. The resulting 32-bit value is then added to the value obtained from S2 [105]

* Important: The actual numerical value of $S_1[105]$ depends on Blowfish standard S-box table.

* The key point is that the first 8-bit portion 01101001, selects entry 105 of S-box 1.

The Blowfish F-function:

$$F(x) = ((S_1[a] + S_2[b]) \bmod 2^{32} \oplus S_3[c]) + S_4[d] \bmod 2^{32}$$

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2) Diffie - Hellman Key Exchange:

Given:

- $g = 5$
- $p = 23$
- Alice's private key $a = 6$
- Bob's private key $b = 15$

Step 1: Alice calculates her public key

$A = g^a \bmod p$

$5^2 = 25 \bmod 23 = 2$

$5^4 = 2^2 = 4$

$5^6 = 5^4 \times 5^2 = 4 \times 2 = 8$

Therefore

Alice's public key $= 8$

Step 2: Bob calculates his public key

$B = g^b \bmod p$

$B = 5^{15} \bmod 23$

Using modular exponentiation:

$5^{15} \bmod 23 = 19$

Step 3: Alice calculate the shared key:

$k = B^a \bmod p$

$k = 19^6 \bmod 23$

$k = 2$

Step 4: Bob calculate the shared key

$k = A^b \bmod p$

$k = 8^{15} \bmod 23$

$k = 2$

Shared secret key $= 2$

3) Blowfish vs RC5 vs DES

Blowfish	RC5	DES
32-448 bits	Variable, commonly 0-2040	56 effective bits
64 bits	Variable, commonly 64 bits	64 bits
Feistel network	Data-dependent rotation	Feistel network
16	Variable, commonly 12	16
Fast in software after key setup	Fast and flexible	Relatively slow by modern
Stronger than DES; legacy today	Flexible and reasonably strong	Considered insecure due to small key size
Large key space and fast encryption	Simple design with variable parameter	Historically important standard.

A) Block cipher vs Stream Cipher.

Feature	Block cipher	Stream Cipher.
Basic operation	Encrypts fixed - size block	Encrypts data one bit at a time
Data processing	Block - by - block	Continuous stream
padding	May require padding	usually does not required padding
Speed	Generally efficient for blocks	very efficient for continuous data
Example	AES	Chacha 20
Scenario	File / database encryption	Real time communication Streaming.

Block cipher example:

* AES can encrypt data in fixed size 128 bit
* Scenario: Encrypting a file stored on a compu

Stream cipher example:

* Chacha20 generate a key stream that is combined with plaintext continuously
* Scenario: Encrypting real time network communication where data arrives continuously

5) Importance of key management in public key crypto system:

* key management is critical because public key cryptography depends on correctly handling public keys, private keys and certificates.

1) Private - key protection:

* A private key must remain secret. If an attacker obtain it, they may decrypt protected information or create fraudulent digital signature.

2) public - key authenticity:

* user need to know that a public key actually belongs to the claimed person or organization.

3) key distribution:

* public key must to distributed securely and reliably.

4) key revocation.

* key compromised or expired keys must be revoked so they cannot continue to be trusted.